

Week - 06 - Graphics\Lecture_11_code_2.py

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1  '''
2  We learn how to move our bomb object on screen with graphics library.
3  In this design, we calculate bomb position on screen by ourself.
4  The coordinate system of our window is different than the real world
5  In graphics library, the x =0, y = 0 is on top-left corner, while is
6  on bottom-left in real world. So we need a translation formula which
7  we develop here
8  '''
9
10 from graphics import *
11 import math
12
13 from Animate import animate
14 from Projectile import *
15
16 class Canon:
17     # Now a composition design
18     def __init__(self, cid, length, type):
19         self.cid = cid
20         self.length = length
21         self.type = type
22         self.bomb = Projectile(201, 25, 12)
23         self.win = GraphWin("Projectile Simulation", 800, 800)
24
25     # def method_name (self, another_object)
26     def fire(self, velocity, theta):
27         x_vel = velocity * math.cos(theta)
28         y_vel = velocity * math.sin(theta)
29         dt = 0.02
30
31         bomb = Circle(Point(10, 390), 5)
32         bomb.setFill('red')
33         bomb.draw(self.win)
34
35         # SCALE represents how many pixels we put in one unit of real-world, for example in
36         # a feet or meter
37         SCALE = 100
38         while self.bomb.y_pos >= 0:
39             dx = x_vel*dt
40             dy = y_vel*dt
41
42             self.bomb.x_pos += x_vel * dt
43             self.bomb.y_pos += y_vel * dt
44             x_vel -= AIR_DRAG * dt
45             y_vel -= G * dt
46
47             print(dx, dy)
48
49             # putting a - sign with y inverts it to represent a real-world like change
50             bomb.move(dx*SCALE, -dy*SCALE)
51
52             time.sleep(0.2)

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52  
53     self.win.getKey()  
54  
55     def load_projectile(self):  
56         self.bomb.x_pos = 0  
57  
58     canon = Canon('C101', 45, 'Vehicle-Mounted')  
59     canon.fire(10, math.radians(60))
```